

Executive Summary: This week the team made significant strides in finishing the WAD final prototype, our device is working and getting all the information and data needed. This means that the last thing to finish is the overall case and documentation of the WAD. We successfully completed field testing for gunshot and GPS accuracy by going out to private property and shooting guns and testing the overall location accuracy while out of city limits. Overall on the hardware side, the assembly, trouble shooting, and final optimization/build of the WAD prototype is completed. For the software side, the documentation is getting completed in the README file and the overall data processing of the WAD. Next week the team plans on finalizing the WAD with all documentation and presentation slides. This means the completion of the Estimated battery runtime/ pcb design. Finishing the Hardware design document and software manual. Then finally completing, polishing, and practicing the Final Demo slides for the end of semester final presentation on Monday May 4th.

Hardware:

Cody C.

Progress:

- 4/22 – Team meeting (~2 Hours 30 Minutes)
 - Uploaded full code to the Arduino Nano ESP32
 - Talked about next steps for testing the Gunshot detection
 - Scheduled field testing for Friday, April 24th at 12pm to test the gunshot detection, as well as how well the GPS works.
 - Measured the current flow between battery and Arduino Nano ESP32
 - 167 mA flowing through.

- Leave the working product on a breadboard but create a working project onto a protoboard.
 - Soldered on components to the protoboard
 - Ryan soldered the Arduino ESP32 to the protoboard and microphone. I soldered the GPS wires to the ESP32 and ran testing by directly connecting the ESP32 to a laptop. Bluetooth works and it receives the GPS and Gunshot decibel readings on the phone.
- Updated Schematics for the CCEW Wearable Device (~20 Minutes)
- Create wiring diagram for the Wearable Device (~20 Minutes)
- Updated GitHub Repo (~50 Minutes)
 - Organize Hardware GitHub Repo
- Get electrical tape to combine the components together
- 4/23 – Finish up soldering in the capstone lab (~ 1 Hour)
 - Solder on the LIPO Charger and connect the pins together and run all testing
 - After some testing, everything works (Bluetooth, GPS, and Microphone), however, the button does not work.
- 4/24 –
 - Troubleshoot button problem and add on LED (~1 Hour)
 - Meet with Ryan to figure out button problem
 - Desolder the wire connected to the button and resolder them back on
 - Field testing for GPS and Gunshot detection with the team (~1 Hours)
 - Gunshot detection:

- Tested with 5 mm pistol and AR-15 rifle
 - Take software readings of the data
 - Run multiple test incrementing further back to test how far the device can go before it can not detect a gunshot
 - Wearable Alert Device was able to detect gunshots majority of the time
- Fix protoboard wiring (~1 Hour 30 Minutes)
 - Fix soldering and wiring of the protoboard and make it compact
 - Move breadboard version into smaller breadboards
- 4/25 –
 - Quick team meeting (~15 Minutes)
 - Talk about presentation
 - Delegate who needs to work on what on the presentation
 - Ryan: GPS, boost converter change
 - Cody: Microphone, microphone change
 - Reorganize smaller breadboard (~30 Minutes)
 - Cut smaller wires for small breadboard and create a compact version
 - Take pictures of the protoboard version and smaller breadboard version

Total Hours Worked: ~ 9 Hours 15 Minutes

Future Work/Plan

- Calculate Estimated runtime
- Start Hardware Documentation

- Start Final Presentation

Hardware:

Ryan K.

Progress:

- 4/22 - Team Meeting (~2 hrs)
 - Went over future plans and important dates
 - Constructed the majority of the protoboard
 - Connected everything but the LIPO charger, GPS, and LED
 - Button could not be picked up by the arduino
- 4/24 - Protoboard Cleanup and Field Testing (~2hrs)
 - Complete protoboard
 - Added remaining components and fixed button issue
 - Tested and recorded real data for the GPS and microphone
- 4/25 - Team Meeting (~15min)
 - Quickly went over the upcoming due dates and had all out Final Demo slides assigned and outlined

Total Hours Worked: 4hr 15min

Future Work/Plan:

- Start and complete all hardware specific documentation
- Start, finish, and practice Final Demo slides with team

Software:

Alex H.

Progress:

- 4/20 (3 h 21 min)
 - Wrote most of the README file for the software development documentation on the GitHub repository, providing details of how to install the Arduino IDE, library dependencies, and how to set up the software development environment.
 - Created a cover page for the user manual.
 - Created two folders in the GitHub repository, one for the documentation and one for the presentations.
- 4/22 (2 h 7 min)
 - Finished writing the README file for the software development documentation.
 - Assisted hardware team members with soldering the final prototype of the Wearable Alert Device.
- 4/24 (50 min)
 - Drove to a volunteer's private property to fire some guns to test the performance of the Wearable Alert Device.
 - A 5 mm pistol and AR-15 rifle were the two firearms used for this test.
 - To collect data, the team screen-recorded a phone that had the nRF Connect mobile app. Simultaneously, the computer screen containing the Serial messages from the Wearable Alert Device was also recorded, and the volunteer firing the guns was also recorded.
 - Made slight adjustments to the Wearable Alert Device's internal settings so that it picks up a gunshot better.
 - **Wearable Alert Device was able to detect a gunshot most of the time.**

- 4/26 (3 h)
 - Converted [README.md](#) file from GitHub repository into a Word doc so that it can be easily accessible for anyone without the need for a GitHub account.
 - Processed data from testing into a nice and presentable format
 - Data will be presented at the final demonstration in the form of a video, where it will show the nRF Connect mobile app, live footage of firing the guns, and a readable table of the data from the mobile app. When the live footage shows a gun being fired, the app and table will update in the video.
 - Began writing nRF Connect app setup section for the user manual.

Total Hours Worked: 9 h 18 min

Future Work/Plan:

- Finish writing the user manual
- Complete final demonstration presentation slides
- Practice final demonstration presentation slides with the team
- Prepare and gather all the project files to store in the USB flash drives.

Project Manager:

Jonas W.

Progress:

- 4/22 – Team meeting (~1 Hours 30 Minutes)
 - Schedule field testing day, work on documentation
 - Worked on the beginnings of the Hardware design Document
 - Worked on the beginnings of the Software User manual

- Work with volunteer to double check date and time for testing of gunshot detection and get everything confirmed
- 4/23 – Presentation (~1 Hours)
 - Create slides, theme and transition for presentation for Final Demo
 - Worked on some of the layout and outline of the overall presentation but needed to talk with team about promised deliverables
- 4/24 – Field testing (~1 Hours)
 - Worked with volunteer at their property to fire guns to get data for thw WAD
 - Fires two guns 1 being a glock 17 pistol and the other an AR-15, **Gunshot detection worked majority of the time**
 - Had to make several adjustments during the testing
 - Took video and screen recordings of data for final presentation
- 4/25 - Mini Team meeting (~30 Minutes)
 - Personally worked on mapping out the outline of the final presentation slides
 - Shared with team what I thought would be the best course of action and made sure everyone was on the same page to end strong
- 4/26 - Self preparation and documentation management (~2 Hours)
 - Worked on more ideas and sketches to begin working on the case of the WAD before taking measurements on Monday 4/27.
 - More refreshing on Autodesk inventor so that once measurements are taken I can create the STL files and have them ready to print this week.
 - Began moving documents over to flashdrives of archive so that when everything begins getting done and I move over to the git hub it will be easier to make the

final movements to the final archive

Total Hours Worked: 6 Hours

Future Work/Plan:

- Take measurements of the Protoboard WAD for case building
- Create and print case for WAD
- Finish final presentation slides
- Practice final presentation
- Finish files and documentations
- Finish file archive